

Menota Lemmatisation Assistant (MLA) 2.0

A User Guide

Odd Einar Haugen, 10 September 2024

NB! This is a detailed user guide. If you would like to get a quick and essential overview of MLA, please go to <https://clarino.uib.no/menota/documentation/mla-documentation>

The first version of MLA was developed in the period 2005–2006 by Paul Meurer at the University Library in Bergen. This second version, also developed by Paul Meurer, is greatly expanded and is no longer just a tool for lemmatisation (and morphological annotation) of Old Norse texts but also an efficient tool for adding a normalised and even a facsimile level to diplomatic texts. Note that there are a few requirements for MLA 2.0 to work:

- A valid XML file according to the Menota schema. An XML editor like Oxygen is recommended, since it offers a continuous checking of the validity of the file.
- MLA is primarily developed for files which contain a diplomatic level of the text and perhaps also a facsimile level, but which lack the normalised level and morphological annotation (for lemma and grammatical forms). The texts from *Gammelnorsk Ordboksverk* in the archive are kept on a single, diplomatic level, while texts from the Menotec project usually contain a facsimile level in addition, but no normalised level.
- Each file must have **xml:id**'s for all words (in the **<w>** element) and punctuation marks (in the **<pc>** element).
- The file must be uploaded to the Menota repository, and the corpus must subsequently be updated (see ch. 7 below).

1. Workflow – getting in and out of MLA

An XML file can be downloaded at any time from MLA and inspected in a suitable XML Editor. This will allow the annotator to check the appearance of the file and to add, delete or revise any part of the encoding. For example, one may want to add or delete **<w>** or **<pc>** in the file. This can be done directly in the XML file. When the file is uploaded and updated, any changes will have full effect.

NB! If a **<w>** or **<pc>** element is added in the XML file, this must be given an **xml:id** attribute. The sequence of **xml:id**'s are in the order of 100, e.g. w100, w200, w300, etc. Any addition should be given in steps of 10, e.g. w210 and w220, if two elements are added between w200 and w300. This allows for large additions. If a **<w>** or **<pc>** element is deleted, there is no need to change the **xml:id**'s of the file. The only consideration is that there cannot be two or more identical **xml:id**'s in the file. The **xml:id**'s are assigned on a file by file basis, so there may be a number of w100 **xml:id**'s in the whole corpus.

Especially when working with a large file, the annotator will download it from MLA and upload & update it a number of times. As will be seen from the guidelines below, the work may be done in batches, for example first normalising the file, then annotating for word class

only, then for the whole range of grammatical categories, and finally, perhaps, adding a facsimile level. Some might like to go “all in” and do the normalisation and morphological annotation in one go.

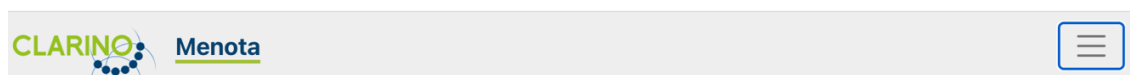
2. Getting in

2.1 Use the browser **Chrome** and open <https://clarino.uib.no/menota>



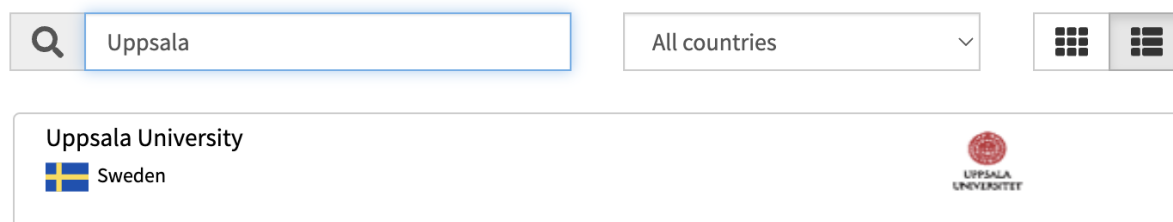
2.2 **Sign in** (to the far right in the menu bar) to access MLA:

On a narrow screen, **Sign in** will be found in a collapsible menu to the right:

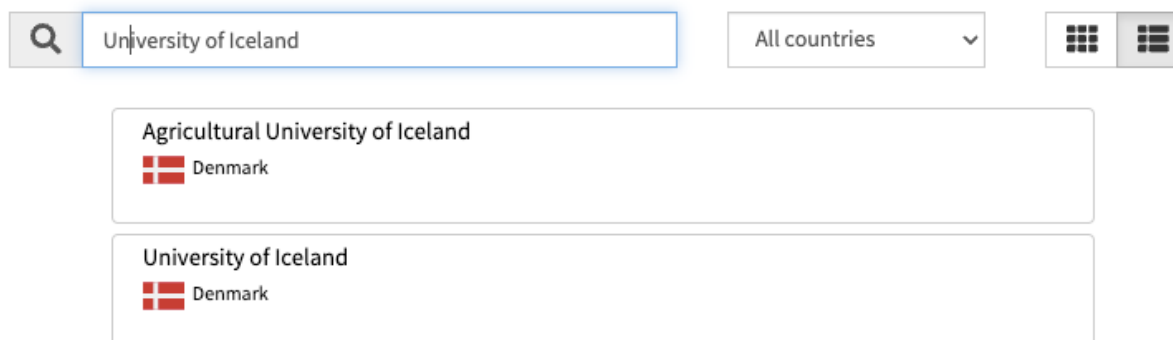


For Norwegian users, **Feide** is probably the simplest option. Under **Tilhørighet** (affiliation) select your affiliated institution, e.g. Universitetet i Bergen. Click on **Fortsett** and select **Bruk arbeids- eller skolekonto**, which will allow you to enter MLA.

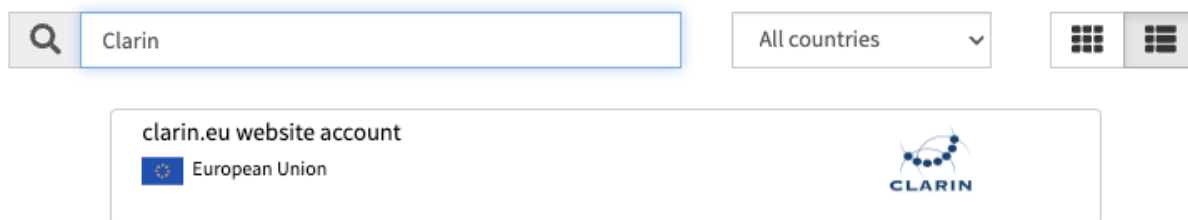
Foreign users must find their own institution on the long list offered:



Icelandic users may be surprised to see a Danish flag, but the link should work anyhow:



If no affiliated institution is accessible, it is possible to register via the Clarin website:



The image shows the Clarin website registration interface. At the top, there is a search bar with a magnifying glass icon and the text 'Clarin'. To the right of the search bar is a dropdown menu labeled 'All countries' with a downward arrow. Further right are two icons: a 3x3 grid and a 2x2 grid. Below these elements is a registration box. On the left, it says 'clarin.eu website account' and 'European Union' with the European Union flag. On the right, there is the CLARIN logo, which consists of a stylized blue and green network of dots and lines.

2.3 You are now in. Click **Corpora** to access one of the three generally accessible corpora in Menota. If it has been agreed with Menota (see ch. 7 below) you may access the **test corpus**, which can be found on <https://clarino.uib.no/menota/catalogue/menota-test>.

Corpus: Test corpus

This is a test catalogue for Menota texts. Texts in this catalogue should not be quoted or referred to in scholarly publications. Texts are displayed on one or more levels: facsimile (very close transcription), diplomatic (less close and with some interpretation), normalised (regularised orthography). Some texts are morphologically annotated and thus allow for more detailed linguistic searches. A search can be performed on a single text or on any combination of texts.

20 texts. Filter: by title/signature by document

Click on a title to go to the text. Click in a row to see the full metadata record.

Signature	MS or Work Title	Main Editor(s)	Annotation	Version	Date	Lang.	Words	Lemma	Form
DN 2:13 test version	1266 : Bergen	Språksamlingane	GNO	0.8.1	1266	nor	164	none	none

Please note that not all texts in the test corpus have been indexed (as required by MLA), so for the time being, these texts cannot be edited in MLA. Contact us if you would like to have a test text indexed (contact details in ch. 7 below).

2.4 Let us have a look at a short text, a Norwegian diploma from 1266, with a total of no more than 164 words, located in the **Test corpus**. This text has been edited, normalised and translated in ch. 5 of *Handbok i norrøn filologi* (Bergen: Fagbokforlaget, 2013, pp. 260–261), in case you would like to have a second opinion on your work. Click on the title in the second column to open it:

DN 2:13 test version	1266 : Bergen	Språksamlingane	GNO	0.8.1	1266	nor	164
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2.5 You are now ready to go, whether you would like to normalise or annotate (or both). These procedures are explained in ch. 4, 5 and 6 below. However, before diving into the text, read ch. 3 on how to navigate in MLA.

Navigation is done almost exclusively with keystrokes and arrow keys, and a minimum of mouse clicks. MLA 2.0 puts great weight on ergonomics.

3. Navigating

3.1 After having opened the selected text, this menu bar will be displayed:

DN 2:13 test version – 1266 : Bergen

Page: < 1r > >| ☒ Dipl ☐ Norm ☐ Clean text

[Bookmarks](#) [Set bookmark](#) [Reset matches](#) [Parse page](#) ☐ POS only [Download XML](#)

Decide which page you will be working on. In a charter, there are usually only two pages, **1r** for the recto (front) page and **1v** for the verso (back) page.

For longer, multi-page texts (typically a codex), you might want to go to a specific page, depending on your work flow. Then, you must return to **Text** in the menu bar and select the next page you would like to edit.

3.2 Click **POS only** (unless you want to do a full annotation) in order to keep the screen less cluttered. Then click **Parse page** which opens the doors to MLA. This screen now appears:

CLARINO **Menota** [Corpora](#) [Text](#) [Search](#) [MLA](#) [Dictionary](#) [Documentation](#) [User guides](#) [Admin](#) Odd Einar Haugen

DN 2:13 test version – 1r

< 1 2 3 4 5 ... 10 >

Go to page:

[Reparse](#)

[Redisambiguate](#)

Show: ☒ Norm ☐ Facs

[Edit normalization](#)

[Edit facs rules](#)

☒ Show used CG rules

☐ Collapse readings

☐ Key shortcuts w/o Ctrl

☒ Show context box

Dipl	Norm	MSA
<pb ed="ms" n="1r"/>		
<lb ed="DN" n="1"/>		
<lb ed="ms" n="1"/>		
Th{orgils}	☐ Thorgils	1 ?? unknown
m{eð}	☐ með	3 með xAP [6058] ...
gvðs	☐ guðs	1 guð xNC [10]
miskun{n}	☐ miskunn	2 miskunn xNC [136] ...
byskop	☐ byskop	1 biskup xNC [15]
i	☐ í	7 í xAP [8913] ...
stawang{re}	☐ stawangri	1 ?? unknown
.	.	1 . xPC
sender	☐ sendir	1 senda xVB [40]
<lb ed="DN" n="2"/>		
lærð om	☐ lærð um	2 lærðr xAJ [1] ...

Begin by finding your point of departure. Within each page there will be a number of screens that follow each other. For a short text like DN 2:13, there are no more than 10 screens, but for larger texts, there will be many, perhaps as many as one for each line of the page. For a charter of 164 words, there will be approx. 15–20 words on each screen. By default, MLA will open with the first screen of the lowest page number.

The meaning of the boxed question mark as well as the proposed lemma in the third column will be explained in ch. 4.1 below.

3.3 Make some initial decisions on how you would prefer to have your screen:

Select **Show used CG rules** if you would like to see the Constraint Grammar rules that are applied. This is probably something you would postpone.

Click **Norm** if you are planning to normalise the text, **Facs** if you are planning to add a facsimile level (probably something you would consider at a later stage, see ch. 5 below).

Click **Collapse readings** if you do not want to see the full list of possible grammatical forms (also something you might want to do at a later stage, see ch. 4 below).

Click **Key shortcuts w/o Ctrl** if you would like as few key combinations as possible (recommended). Many commands can then be done by pressing a single key, rather than the **ctrl** key + a specific key.

Click **Show context box** if you want to have the full text visible in the right half of the screen. Try this option and see what you prefer.

3.4 Start moving around

Move from one screen to the next by **ctrl-shift-arrow-right** and to the previous one by **ctrl-shift-arrow-left**. For these commands, the **ctrl**-key must always be used.

Put the cursor in one of the lines. Use the **arrow** keys to move **upwards** or **downwards** on the screen.

3.5 Start interacting with the text. Select a line, for example the first one:

Th{orgils}	 Thorgils	1 ?? unknown
------------	--	--------------

Click **m** (or **ctrl-m** if you have not ticked the box for **Key shortcuts w/o Ctrl**), and you will be able to edit the normalised form in a box that now appears. More on this in ch. 4 below.

Click **arrow-right** (or **ctrl-arrow-right**) to be able to edit the lemma and grammatical information of the word. More on this in ch. 6 below.

At this point, you might like to make comments (notes) to yourself, so that you can return to a specific place later on. Click **ctrl-k** to open a comment box, **ctrl-s** to save the comment, and **escape** to discard changes. See the list of keyboard shortcuts in Appendix 2 below.

3.6 Change the delimitation of words. MLA will import the **<w>** and **<pc>** elements from the file, displaying them line by line. For normalisation as well as lemmatisation, it is sometimes necessary to join or split **<w>** elements. This can be done from inside the MLA application.

3.6.1 Joining words. The words “salo” and “hjalpar” should be joined into a single word “salohjalpar”, as most dictionaries would prescribe (see e.g. “sáluhjǫlp” in ONP):

salo	☐ salu	3	sála	xNC	[27] ...
hjalpar	☐ hjalpar	2	hjǫlp	xNC	[24] ...

Select the first word, then click **j** (or **ctrl-j**) for “join”, and MLA will do its magic, leaving a small joining symbol in red:

salo hjalpar	✚	☐ salo hjalpar	1	sáluhjǫlp	xNC	[2]
--------------	---	---------------------------------------	---	-----------	-----	-----

The two words “salo hjalpar” have now been joined into a single word (although with an internal space), and this can be normalised and annotated as a single word. More on these procedures in ch. 4 and 6 below.

This is what the word looks like in the XML file which MLA will export:

```
<w xml:id="w9700">
  <choice>
    <me:dipl>salo hjalpar</me:dipl>
    <me:facs>&slong;alo &hrdes;&inodot;alpar</me:facs>
    <me:norm>salo hjalpar</me:norm>
  </choice>
</w>
```

The normalised form “salo hjalpar” is not correct; it should be “sáluhjǫlp”. We will return to this in ch. 4 below.

Ouch! You have joined two words which should not have been joined. Select the word, then type **j** (or **ctrl-j**) again, and the two words are re-instated.

3.6.2 Splitting words. The two words “viðr komandum” are encoded with a space between them within a single **<w>** element:

viðr komandom	☐ viðr komandum	1	viðrkomandi	xAJ	[2]
---------------	--	---	-------------	-----	-----

In order to split them into “viðr” and “komandum”, select the line and click **y** (or **ctrl-y**). Think of **y** as a character that is split into two parts. This screen now appears:

Split word



... <pb/> <lb/> <lb/> Th{orgils} m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb> lærð|om </lb>
 {ok} lendum verandom {ok} **viðr komandom** ollvm gvðs vínun {ok} <lb/> sínun <lb/> þæim sem
 þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollum <lb/> monnom kunnikt gera at
 ...

Word: **viðr komandom**

Facs: vıð&rrot; koman&drot;om

Dipl: viðr komandom

Cancel

OK

Add a # between the two words. In this case the # will replace the space, if the word to be split does not have an internal space, just add the # where it should be split:

Split word



... <pb/> <lb/> <lb/> Th{orgils} m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb> lærð|om </lb>
 {ok} lendum verandom {ok} **viðr komandom** ollvm gvðs vínun {ok} <lb/> sínun <lb/> þæim sem
 þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollum <lb/> monnom kunnikt gera at
 ...

Word: **viðr komandom**

Facs: vıð&rrot;#koman&drot;om

Dipl: viðr#komandom

Cancel

OK

After having clicked **OK**, the line has been split in two, and there is a small splitting symbol in red as a reminder. The two words can now be edited separately:

```
<seg>
viðr ↵ [?] viðr 7 viðr xAP [165] ...
komandom [?] komandum 1 ?? unknown
</seg>
```

In the XML text exported from MLA, the split words will be encoded according to the rules of Menota (cf. ch. 5.3.2). Note that MLA automatically updates the **xml:id** of the second word:

```

<seg type="nb">
  <w xml:id="w1500">
    <choice>
      <me:dipl>viðr</me:dipl>
      <me:facs>v&inodot;ð&rrot;</me:facs>
      <me:norm>viðr</me:norm>
    </choice>
  </w>
  <w xml:id="w1510">
    <choice>
      <me:facs>koman&drot;om</me:facs>
      <me:dipl>komandom</me:dipl>
      <me:norm></me:norm>
    </choice>
  </w>
</seg>

```

There is no normalised form of the second word. We will deal with this in ch. 4 below.

Ouch! You have split a word which should not have been split. With the first line selected, type **y** (or **ctrl-y**) again, and you are offered the possibility of undoing the splitting:

Undo split word
×

... <pb/> <lb/> <lb/> Th{orgils} m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb>lærð|om </lb> {ok} lendom verandom {ok} <seg>viðr komandom </seg> ollvm gvðs vínun {ok} <lb/> sínum <lb/> þeim sem þetta bræf sia eða hœyra Quediu Guðs ok sína . ver vilium ollum <lb/> monnom kunnikt gera ...

Undo splitting of: viðr#komandom

Cancel OK

Click **OK** and you are back on track.

Finally, disregard any elements in a field when editing a word. Here, the **<lb>** element states that there is a linebreak within the word:

```
far<lb n="9" ed="DN"/>sældar
```

Do not change anything here, just insert the **#** where you would like to split the word:

```
far<lb n="9" ed="DN"/>#sældar
```

Ouch (again)! It was not a good idea to split this word (it really is a single word), so you may retrace your steps as explained above.

4. Normalising and lemmatising a text

4.1 After having opened the selected text, in this case **DN 2:13 test version**, and having ticked **POS only**, the screen will open with three columns.

Note that it is only possible to parse foliated pages (**1r**, **1v**, ...), not pages entitled **abstracts**, **image**, **translation** or the like.

The first column is the text on a diplomatic level, in which expanded abbreviations are displayed in curly brackets, e.g. **han{n}**, and uncertain readings in bottom half brackets, e.g. **h_lo_jn**.

The second column contains a proposed normalisation.

The third column offers a proposed lemma and its word class (see the *Menota Handbook* ch. 11.6.3 for a recommended list of word classes). The proposed lemmata are based on the whole Menota corpus and are surprisingly often correct, although with a notable exception for proper names.

Th{orgils}	☐ Thorgils	1 ?? unknown
m{eð}	☐ með	3 með xAP [6058] ...
gvðs	☐ guðs	1 guð xNC [10]
miskun{n}	☐ miskunn	2 miskunn xNC [136] ...

A small number at the opening of the third column indicates the number of alternatives.

4.2 Begin by normalising the text, which is done by editing the second column. If the proposed normalised form is correct, type **s** (or **ctrl-s**) in the selected line, and a check mark will replace the question mark, indicating that the normalisation has been accepted. Move from one line to another with the **arrow-down** (or **arrow-up**) key.

Here, the normalisation of the second, third and fourth word have been accepted and saved to the MLA file:

Th{orgils}	☒ Thorgils	1 ?? unknown
m{eð}	✓ með	3 með xAP [6058] ...
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

For the first word, you would probably like to have “Porgils” rather than “Thorgils”. Select this line and type **m** (or **ctrl-m**) to open an edit window:

Add norm/facs



... <pb/> <lb/> <lb/> **Th{orgils}** m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb> lærð|om </lb>
 {ok} lendum verandom {ok} <seg> viðr komandom </seg> ollvm gvðs vnum {ok} <lb/> sínum <lb/>
 þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollum <lb/> monnom
 kunnikt gera ...

Dipl: **Th{orgils}**

Dipl XML: **Th<ex>orgils</ex>**

Norm XML:

Cancel

OK

Type “Porgils” in the open field. There are no proposed forms here, but in many cases there will be two or more alternatives, which you can select simply by clicking. Remember to press **OK** when leaving the window.

Th{orgils}	✓ Porgils	1 ?? unknown
m{eð}	✓ með	3 með xAP [6058] ...
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

4.3 Continue (if you like) by lemmatising the text, i.e. adding or correcting the lemma (dictionary entry) as well as the word class (part of speech, **POS**).

The four initial words of the charter have now been normalised, but the lemma and word class are missing (or might be wrong) for the first word. Having selected this line, press **arrow-right** (or **ctrl-arrow-right**) in order to select the word for annotation. Press **m** (or **ctrl-m**) in order to open a new edit window:

Add reading



... <pb/> <db/> <db/> **Th{orgils}** m{eð} gvðs miskun{n} byskop i stawang{re} . sender <db> lærd{om} </lb>
 {ok} lendom verandom {ok} <seg> viðr komandom </seg> ollvm gvðs vínun {ok} <db/> sínum <db/>
 þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollum <db/> monnom
 kunnikt gera ...

Dipl: **Th{orgils}**

Dipl XML: **Th<ex>orgils</ex>**

Norm XML:

Lemma:

☐ empty lemma and features

Features:

POS: **xVB** verb · **xNC** noun, common · **xNP** noun, proper · **xAJ** adjective · **xAV** adverb, general ·
xAQ adverb, interrogative · **xAP** preposition · **xCC** conjunction · **xCS** subjunction ·
xRP relative pronoun · **xPE** pronoun, personal · **xPR** pronoun, reflexive ·
xPQ pronoun, interrogative · **xPI** pronoun, indefinite · **xDD** determiner, demonstrative ·
xDO determiner, quantifier · **xDP** determiner, possessive · **xNA** numeral, cardinal ·
xNO numeral, ordinal · **xIM** infinitive marker · **xVP** verbal particle · **xIT** interjection ·
xPD pronoun, general · **xNU** numeral, general · **xCU** conjunction, general · **xAT** article ·
xNE enclitic · **xEX** expletive particle · **xFW** foreign word · **xFN** foreign name · **xUA** unassigned

Cancel

OK

The normalised form “Þorgils” has been saved, as stated in the **Norm XML** field, but you have to add the lemma. That would be the nominative case of the word, which is identical with the normalised form, “Þorgils”. However, we suggest that you follow ONP and use an initial small letter, “þorgils”.

The next step is to assign word class, which is **xNP** (proper noun). So this is what the edit window should look like:

Add reading



... <pb/> <db/> <db/> **Th{orgils}** m{eð} gvðs miskun{n} byskop i stawang{re} . sender <db> lærd{om} </lb>
 {ok} lendom verandom {ok} <seg> viðr komandom </seg> ollvm gvðs vínun {ok} <db/> sínum <db/>
 þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollum <db/> monnom
 kunnikt gera ...

Dipl: **Th{orgils}**

Dipl XML: **Th<ex>orgils</ex>**

Norm XML:

Lemma:

☐ empty lemma and features

Features: xNP

POS: **xNP** noun, proper

Cancel

OK

Roman numerals should not be lemmatised, so select the word class **xEY** (for ‘empty’). We also recommend that they are encoded with the attribute **xml:lang="lat"**. See the *Menota Handbook* ch. 11.6.3 on Roman numerals.

The same applies to punctuation characters, which are encoded in the **<pc>** element. Select **xEY** as lemma also for these.

The first word now appears with a check mark against the normalised form as well as against the lemma and its word class.

Th{orgils}	✓ Þorgils	1 ✓ þorgils xNP
m{eð}	✓ með	3 með xAP [6058] ...
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

Save the proposed lemmata (with word classes) for the three remaining words. If necessary, select the third column by **arrow-right** (or **ctrl-arrow-right**), as seen here. For the second word, there are three possible forms, indicated by the number in front of the proposed lemma. The spelling of the word is the same, but it can be assigned to three different word classes, **xAP**, **xVP** and **xAV**. By far, the first word class is the most frequent one (and this is why it has been listed first), having 6058 instances in the whole Menota corpus, while the two next ones are far less common.

Th{orgils}	✓ Þorgils	1 ✓ þorgils xNP
m{eð}	✓ með	3 með xAP [6058] með xVP [109] með xAV [57]
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

Texts in the Menota archive have been annotated in various project with varying rules. In the *Menota Handbook*, ch. 11.6.3 we suggest stricter guidelines in this field. Based on these recommendations, we go for **xAP** (preposition) as the word class, not **xVP** (verbal particle) or **xAV** (adverb). In order to select the form, type **ctrl-enter**. Having done this for the third and fourth word, too, we are told by MLA that the text has been normalised according to the second column and lemmatised according to the third column.

Th{orgils}	✓ Þorgils	1 ✓ þorgils xNP
m{eð}	✓ með	1/3 ✓ með xAP [6058]
gvðs	✓ guðs	1 ✓ guð xNC [10]
miskun{n}	✓ miskunn	1/2 ✓ miskunn xNC [136]

4.4 The proposed normalisation of the text is based on a number of rules developed in July and August 2023. They cover a number of cases, but there will certainly be rules that you might like to add for recurring words. This can be done on a word-by-word basis pressing the **Edit normalization** button.

For example, the form “ollvm” should be normalised to “øllum”, “gvðs” to “guðs”, and “vínvm” to “vínvm”.

Edit normalization rules

×

Add rule

dipl: ollvm

norm: øllum

Add

Apply

Add dipl form as 'word' for whole word, 'prefix-' for prefix, '-suffix' for suffix, '-int-' for internal string, '+nonini-' for non-initial string. Add norm replacement without hyphens or plus sign. See rules below for examples.

Rules

Dipl	Norm	Remove
a	á	🗑
adr	áðr	🗑
aðr	áðr	🗑

Click **Add** and (if you like) check that the rule has been added:

Nu	Nú	🗑
Nv	Nú	🗑
ollafr	Óláfr	🗑
Ollum	Øllum	🗑
ollvm	øllum	🗑
Ollvm	Øllum	🗑
or	ór	🗑

Then click **Apply** to make it work, and continue with the two next words in the same way.

Have a look at the explanation below the **Apply** button. It is possible to normalise parts of a word (initial, medial, final) and to specify exceptions from the rule, e.g. that ‘þ’ should be normalised to ‘ð’, apart from in the beginning of a word.

If you think that a rule is wrong or that it is over-generating, you may click the wastebasket button in order to delete it.

After having added new normalisation rules, MLA will become even better at suggesting normalised forms.

4.5 Punctuation is encoded by the **<pc>** element. When normalising a text, the punctuation usually has to be adjusted considerably. While some codices have too many punctuation marks compared to a normalised text, charters frequently are in need of additional punctuation. We recommend editing the punctuation in the same workflow as the running text, line by line, since whenever a dot (i.e. a full stop) is encoded on the normalised level, the next word should have a capital letter. Note that we do not make any distinction between dots in manuscripts, which have a rather indiscriminate usage, and the full stops in a normalised text. Both are encoded by the same character, U+002E FULL STOP.

4.5.1 Editing a punctuation character. Punctuation characters (as well as any words) will be signalled by a boxed question mark if they have not been normalised. This is an example from *Elíss saga ok Rósamundar* in Upps DG 4–7 fol, in which the words have been normalised (cf. the check marks), but so far not the punctuation character.

Here, you might want to change the punctuation character from a dot (i.e. a full stop) to a colon:

{oc}	✓ ok	1/10 ✓ ok xCC [15826] <i>select:38</i>
mællti	✓ mælti	1/2 ✓ mæla xVB [861]
<lb n="2" ed="ms"/>		
han{n}	✓ hann	1/2 ✓ hann xPE [17451]
þa	✓ þá	1/4 ✓ þá xAV [11707]
til	✓ til	1/5 ✓ til xAP [10343]
þæirra	✓ þeirra	1/2 ✓ sá xDD [213]
með	✓ með	1/3 ✓ með xAP [6083]
astsamligu{m}	✓ ástsamligum	1 ✓ ástsamligr xAJ
orðum	✓ orðum	1 ✓ orð xNC [172]
.	? .	1 . xPC

Punctuation characters are edited in the same manner as words (see 4.2 above). Select the line with the punctuation character and open the edit window by pressing **m** (or **ctrl-m**) and make a change from dot to colon in the **Norm XML** field:

... . <seg> í millom </seg> ríkiá <lb> lendrri manna </lb> sinna . oc annarra haufðingia . er hann hafðe þangat <cb> <lb> stefnt til rikrar væitzlu . þæirrar hatiðar . {oc} mællti <lb> han{n} þa til þæirra með astsamligu{m} orðum . </p> </div> <div> <head> <supplied/> </head> <p> <lb> Lyðet h{er}rar kuað h{ann} røðu min{n}i . oc gefit hæil <lb> oc holl rað bæiðingum mínum . set huat bazt berr . <lb> oc samir . oc ...

Dipl: .

Dipl XML: .

Norm XML: :

☐ empty norm

Afterwards, MLA states that there is a dot on the **dipl** level (first column), as it should, but a colon on the **norm** level (second column).

með	✓ með	1/3 ✓ með xAP [6083]
astsamligu{m}	✓ ástsamligum	1 ✓ ástsamligr xAJ
orðum	✓ orðum	1 ✓ orð xNC [172]
.	✓ :	1 . xPC

4.5.2 Deleting a punctuation character. Still in *Elíss saga*, we would probably delete the punctuation character right before the subjunction “at” (unless you follow German or older Danish rules and always add a comma at the opening of a clause):

með	✓ með	1/3 ✓ með xAP [6083]
sua	✓ svá	1/3 ✓ svá xAV [1573]
naturu ligre	✓ náttúruligri	1 ✓ náttúruligr xAJ
goðgirnd	✓ góðgirnd	1 ✓ góðgirnd xNC
.	☐ .	1 . xPC
at	✓ at	1/14 ✓ at xCS [8237]
alldregi	✓ aldrigi	1/2 ✓ aldrigi xAV [54]
<lb n="26" ed="ms"/>		
var	✓ var	1/10 ✓ vera xVB [5737]
mær	✓ mærr	1 ✓ mærr xNC [8]

Select the line with the punctuation mark, press **m** (or **ctrl-m**) and select **xEY** as the word class.

... Meðan hann var ungr oc a æsku alldre Ø <lb/> þa var hann fullnomi allra goðra siða . oc
 hirðlegs <lb> hav{er}sc|æiks </lb> . með sua naturu ligre goðgirnd . at alldregi <lb/> var mæer ne
 eckia . ne hialplausir omagar . ren{n}tir <lb/> sin{n}a soemða . æigna ne erfða . ne fiarluta . af hans
 <lb/> raðum ne rangendom . Helldr var iamnan hugr ...

Dipl: .

Dipl XML: .

Norm XML: ☒ empty norm

The punctuation mark will be kept on the diplomatic (and facsimile) level, but on the normalised level, there will be no mark.

Afterwards, the zero symbol will be displayed in the second column stating that there should be no punctuation character on the normalised level, while the dot correctly is kept on the dipl level (first column):

með	✓ með	1/3 ✓ með xAP [6083] <i>select:31</i>
sua	✓ svá	1/3 ✓ svá xAV [1573] <i>select:31</i>
naturu ligre	✓ náttúruligri	1 ✓ náttúruligr xAJ [1]
goðgirnd	✓ góðgirnd	1 ✓ góðgirnd xNC [1]
.	✓ Ø	1 . xPC
at	✓ at	1/14 ✓ at xCS [8237] <i>select:31</i>
alldregi	✓ aldrigi	1/2 ✓ aldrigi xAV [54] <i>select:31</i>
<lb n="26" ed="ms"/>		
var	✓ var	1/10 ✓ vera xVB [5737] <i>select:31</i>
mæer	✓ mæer	1 ✓ mæer xNC [8]

4.5.3 Adding a punctuation character. If you would like to add a punctuation character, select the line right above the planned punctuation character and press **p** (or **ctrl-p**). In this example, you might want to add a comma after “alldre” (at least if you follow modern Norwegian punctuation rules):

Meðan	✓ Meðan	1/3 ✓ meðan xCS [3] <i>select:31</i>
han{n}	✓ hann	1/2 ✓ hann xPE [17451] <i>select:31</i>
var	✓ var	1/10 ✓ vera xVB [5737] <i>select:31</i>
ungr	✓ ungr	1 ✓ ungr xAJ [25]
{oc}	✓ ok	1/10 ✓ ok xCC [15826] <i>select:31</i>
a	✓ á	1/19 ✓ á xAP [6388] <i>select:31</i>
æsku	✓ øsku	1 ✓ øska xNC [6]
alldre	✓ aldri	1/3 ✓ aldr xNC [29] <i>select:31</i>
<lb n="24" ed="ms"/>		
þa	[?] þá	4 þá xAV [11707] ...
var	[?] var	10 vera xVB [5737] ...
h{ann}	[?] hann	2 hann xPE [17451] ...

A new window opens in which you can add a punctuation character in the **Norm** field. The default character is the dot (as shown here), but in this case, it should be replaced by a comma. Note that the **Dipl** field should be left empty, since there is no dot at this level.

A new punctuation character will now be added by MLA, indicated by two symbols. The first symbol, the empty character, states that there is no punctuation character on the diplomatic level, and the next symbol, a red square, that an addition has been made. On the normalised level, a comma is now displayed:

a	✓ á	1/19 ✓ á xAP [6388] <i>select:31</i>
æsku	✓ øsku	1 ✓ øska xNC [6]
alldre	✓ aldri	1/3 ✓ aldr xNC [29] <i>select:31</i>
<lb n="24" ed="ms"/>		
þa	[?] þá	4 þá xAV [11707] ...
var	[?] var	10 vera xVB [5737] ...
h{ann}	[?] hann	2 hann xPE [17451] ...

Ouch! You realise that you have added a punctuation mark by mistake. Select the line above the punctuation mark, then type **p** (or **ctrl-p**) again, and the mark will disappear.

4.6 At this point you might want to check the normalisation and annotation in the XML file.

Return to **Text** in the menu bar, and click on **Download XML**:

The screenshot shows the CLARINO Menota web interface. The top navigation bar includes links for Corpora, Text, Search, MLA, Dictionary, Documentation, User guides, and Admin. A user profile for Odd Einar Haugen is visible on the right. Below the navigation bar, the text 'DN 2:13 test version – 1266 : Bergen' is displayed. A pagination control shows 'Page: 1r' with navigation arrows. Below this are several buttons: 'Bookmarks', 'Set bookmark', 'Reset matches', 'Parse page', 'POS only' (with a checked checkbox), and 'Download XML'.

Depending on your browser, you can open the XML file directly from the downloaded files on your computer, and be reassured that the four initial words in the charter DN 2:13 have received the correct normalisation and lemmatisation (so far, the **facs** level is empty):

```
<pb ed="ms" n="1r"/>
<lb ed="ms" n="1"/>
<lb ed="DN" n="1"/>
<w lemma="þorgils" me:msa="xNP" xml:id="w100">
  <choice>
    <me:dipl>Th<ex>orgils</ex></me:dipl>
    <me:facs></me:facs>
    <me:norm>þorgils</me:norm>
  </choice>
</w>
<w lemma="með" me:msa="xAP" xml:id="w200">
  <choice>
    <me:dipl>m<ex>eð</ex></me:dipl>
    <me:facs></me:facs>
    <me:norm>með</me:norm>
  </choice>
</w>
<w lemma="guð" me:msa="xNC" xml:id="w300">
  <choice>
    <me:dipl>gvðs</me:dipl>
    <me:facs></me:facs>
    <me:norm>guðs</me:norm>
  </choice>
</w>
<w lemma="miskunn" me:msa="xNC" xml:id="w400">
  <choice>
    <me:dipl>miskun<ex>n</ex></me:dipl>
    <me:facs></me:facs>
    <me:norm>miskunn</me:norm>
  </choice>
</w>
```

Remember that in order to have the normalised level displayed, the header must specify that the charter has been encoded on a diplomatic as well as a normalised level:

```
<normalization me:level="dipl norm">
  <p>This text has been encoded on a diplomatic and normalised level.</p>
</normalization>
```

The same applies for any addition of a facsimile level. The **<normalization>** element must be edited manually in the XML file.

4.7 One column at the time? First of all, it should be emphasised that MLA can be used solely for normalisation, in which case you would edit the first column only.

Even if the plan is to both normalise and lemmatise, it can be quicker to do one operation at a time. If the proposed normalised forms are to a high degree correct you may prefer to move down line by line and only build a normalised text. As the next operation, the lemmatisation may be done, also going down line by line.

This is a matter of personal preference. Some would like to do the full morphological annotation, too. More on this in ch. 6 below.

5. Adding a facsimile level

5.1 If you are planning on making a multi-level encoding of a file, it is highly recommendable to begin with the facsimile level and then add the diplomatic and possibly the normalised level afterwards. This has been the procedure for the law texts in the Menotec project, which have been encoded on a facsimile and diplomatic level.

However, if you would like to backtrack, as it were, it is possible to add a facsimile level to a file which only has the diplomatic level. In texts with a low degree of abbreviation, such as in most Norwegian (and Swedish and Danish) manuscripts, there is a limited number of conversions to be made. In many Icelandic texts, on the other hand, there is a huge amount of abbreviations. For these texts, adding a facsimile level is daunting.

5.2 Begin by ticking the box for **Facs** after having pressed **Parse page**. You may or may not tick the **Norm** box.

DN 2:13 test version – 1r

< 1 2 3 4 5 ... 10 > Go to page: Reparse Redisambiguate | Show: ☒ Norm

☒ Facs Edit normalization Edit facs rules ☒ Show used CG rules ☒ Collapse readings

☒ Key shortcuts w/o Ctrl ☐ Show context box

The page which now opens offer proposed facsimile readings based on a list of facs rules. They can be edited in the same manner as the normalisation rules:

Rules

Dipl	Facs	Remove
-d-	&drot;	
-dr-	&drot;&rrot;	
-ðr-	ð&rrot;	
-<ex>an</ex>-	<am>&bar;</am>	
-<ex>ann</ex>-	<am>&bar;</am>	
-<ex>er</ex>-	<am>&er;</am>	
-<ex>ir</ex>-	<am>&er;</am>	

There are basically two types of facs rules. One concerns **character variants**, e.g. that an “s” in a diplomatic rendering should be replaced by the tall “ſ” in a facsimile rendering (if indeed this is the case in the manuscript). Another concerns **abbreviations**, e.g. that “ok” (or “oc”) in a diplomatic rendering should be replaced by the Tironian note “7”.

It will soon be clear that the facs rules are much more dependent on the particular manuscript than the normalisation rules. One should probably begin by analysing the orthography of the manuscript and set up the likely facs rules before beginning the work in MLA. It is essential to have a facsimile of the manuscript in high resolution so as to check the exact orthography and system of abbreviations.

The first words of DN 2:13 tells us that much work remains to be done on the facsimilation of the text:

Dipl	Norm	Facs	MSA
<pb ed="ms" n="1r"/>			
<lb ed="DN" n="1"/>			
<lb ed="ms" n="1"/>			
Th{orgils}	✓ Þorgils	[?] Th{o2g1lf	1 ✓ þorgils xNP [1]
m{eð}	✓ með	[?] m{eð	1/3 ✓ með xAP [6058] <i>select:31</i>
gvðs	✓ guðs	[?] gvðf	1 ✓ guð xNC [10]
miskun{n}	✓ miskunn	[?] m1fkun̄o	1/2 ✓ miskunn xNC [136] <i>select:31</i>
byskop	✓ byskup	[?] býfkop	1 ✓ biskup xNC [15]
i	✓ í	[?] i	1/7 ✓ í xAP [8913]
stawang{re}	✓ Stafangri	[?] ftawang{re>	1 ✓ stafangr xNP

5.3 At the moment, the facsimilation of a diplomatic text is in a test phase. Hopefully, further work on the facs rules will lead to a much higher success rate.

6. Full morphological annotation

6.1 So far, we have restricted ourselves to POS only, so that the morphological annotation only includes the lemma (citation form in a dictionary) and its word class (part-of-speech)

The next and final step is to include all grammatical forms for inflected words. For non-inflected words such as prepositions, conjunctions and subjunctions, there is nothing more to add. For inflected word classes, there is a number of categories to fill in, especially for verbs, and with respect to the high degree of syncretism, also for the weak declension of adjectives.

The annotation procedure is the same as described in ch. 4 above, but more time must be spent on the selection of categories and features for a full morphological annotation. The grammatical categories and their abbreviations are explained in the *Menota Handbook* ch. 11.5 and 11.6, and also in the appendix at the end of this document.

Now, the button **POS only** must not be pressed (nor the **Facs** box discussed in the previous chapter). At this point you might on the other hand tick the box **Collapse readings** in order to have a less cluttered screen.

Let us go to the next page in the charter DN 2:13 and have a look at the phrase “þeim sem þetta bréf sjá eða heyra”. This is what the screen looks like with collapsed readings, i.e. just one morphological analysis for each line:

þæím	✓ þeim	3	sá	xDD nP gU cD [24] ...
sem	✓ sem	11	sem	xCS [1703] ...
þetta	✓ þetta	8	sjá	xDD nS gN cA [574] ...
bræf	✓ bréf	4	bréf	xNC nP gN cA sI [24] ...
sia	✓ sjá	22	sjá	xVB tPS fI vA [100] ...
eða	✓ eða	1	eða	xCC [2039]
hœyra	✓ heyra	5	heyra	xVB tPS fI vA [15] ...

The checkmarks in the second column tell us that this part of the text has been normalised (no more boxed question marks), and the lack of check marks in the third column means that we need to select the correct analysis here.

Well, what do you know! The proposed lemmata in blue are all correct. And the grammatical analysis, too. How come? MLA draws upon the whole annotated Menota corpus, which (if I have counted correctly) amounts to approx. 850.000 words. So the great majority of diplomatic forms have already been annotated somewhere.

However, note that apart from the conjunction “eða” there are more than one alternative for each word. For “þeim” there are three alternatives. After having selected this line, press **arrow-right**, and the whole set opens:

Here, the conclusion is simple: the first analysis is the best one, and as can be seen from the number at the end, also the most frequently chosen one.

6.3 Fine! Let us hunt for a mistake by MLA. This we find two words below.

bræf	✓ bréf	4	bréf xNC nP gN cA sI [24]
			bréf xNC nS gN cA sI [17]
			bréf xNC nP gN cN sI [14]
			bréf xNC nS gN cN sI [1]

It is indeed true that the lemma is “bréf”, and that the case (based on the context) is accusative, **cA** (since it is part of a direct object), but it is in the singular, **nS**, not in the plural, **nP**. So we should go for the second analysis offered. Move down to this line by the arrow key, then press **ctrl-return**, and it has been selected:

þetta	✓ þetta	8	sjá xDD nS gN cA [574] ...
bræf	✓ bréf	1/4	✓ bréf xNC nS gN cA sI [17]
sia	✓ sjá	22	sjá xVB tPS fI vA [100] ...

Have a look at the word above and note that the adjective and the noun have received a similar analysis, **nS gN cA**. A good sign.

6.4 What about words for which MLA have no proposed lemma and grammatical analysis, like “hiauero” on the third page?

hiauero	☐ hjaueru	1	?? unknown
---------	-----------	---	------------

We might begin by normalising the word, which must be an oblique form, “hjáveru”, of the noun “hjáværa”:

hiauero	✓ hjáveru	1	?? unknown
---------	-----------	---	------------

If in doubt, look up the word in ONP – and be pleasantly surprised to see that it has been excerpted by the dictionary:

hjá·vera sb. f.

✓ excerpted ✓ citation slips citation text supplemented structured definitions in Danish definitions in English

Article *Comp., Gloss., Litt., &c.*

Citations by ms. date Citations by source

Search/filter 19 citations...

(see citation slip for text)	•NoDipl (1266) 40 ²⁰
með raðe ok <u>hiauero</u> vars virðulegs herra	•NoDipl (1266) 40 ²⁰

From ONP you will note that this seems to be a typical word for Norwegian charters (DN).

Selecting the third column by **arrow-right**, we have to do a bit of manual annotation. Press **m** (or **ctrl-m**) in order to open the **Add reading** window. Begin by typing the lemma “hjávera” in the lemma window (initially filled by question marks).

Dipl XML: **hiauero**

Norm XML:

Lemma: ☐ empty lemma and features

Features:

POS: **xVB** verb · **xNC** noun, common · **xNP** noun, proper · **xAJ** adjective · **xAV** adverb, general · **xAQ** adverb, interrogative · **xAP** preposition · **xCC** conjunction · **xCS** subjunction · **xRP** relative pronoun · **xPE** pronoun, personal · **xPR** pronoun, reflexive · **xPQ** pronoun, interrogative · **xPI** pronoun, indefinite · **xDD** determiner, demonstrative · **xDQ** determiner, quantifier · **xDP** determiner, possessive · **xNA** numeral, cardinal · **xNO** numeral, ordinal · **xIM** infinitive marker · **xVP** verbal particle · **xIT** interjection · **xPD** pronoun, general · **xNU** numeral, general · **xCU** conjunction, general · **xAT** article · **xNE** enclitic · **xEX** expletive particle · **xFW** foreign word · **xFN** foreign name · **xUA** unassigned

Now, move down to features and pick the correct grammatical form. Having selected xNC for common noun, all features relevant for this word class are displayed:

Features: xNC

POS: xNC noun, common

Number: nS singular · nP plural · nD dual · nSP singular or plural · nU unspecified number

Gender: gM masculine · gF feminine · gN neuter · gMF masculine or feminine · gMN masculine or neuter · gFN feminine or neuter · gMFN masculine or feminine or neuter · gU unspecified gender

Case: cN nominative · cA accusative · cD dative · cG genitive · cAD accusative or dative · cAG accusative or genitive · cGD genitive or dative · cAN accusative or nominative · cO oblique (i.e. non-nominative) · cB ablative · cV vocative · cU unspecified case

Definiteness: sI indefinite · sD definite · sU unspecified definiteness

Move downwards and click on the correct features. This is a fairly quick process (unless you are not sure about the analysis). This should be the result:

Features: xNC nS gF cD sI

POS: xNC noun, common

Number: nS singular

Gender: gF feminine

Case: cD dative

Definiteness: sI indefinite

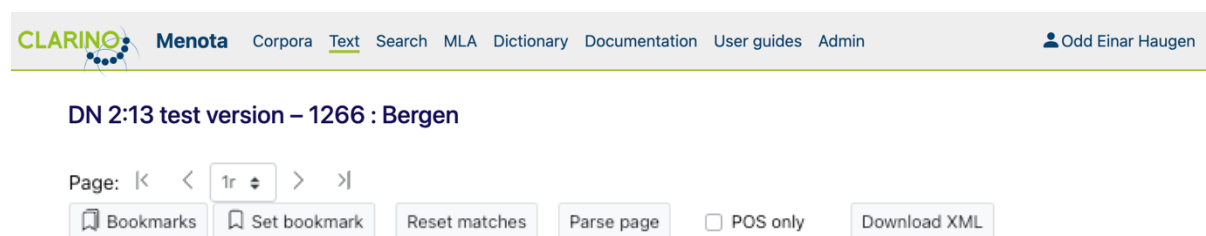
Click **OK** when you are satisfied with the lemma and its grammatical form (you may change anything within this window), and see that the word now has received a full morphological annotation:

hiauero ✓ hjáveru 1 ✓ **hjávera** xNC nS gF cD sI

A particular challenge are proper nouns, whether of once living persons or of fictive ones, place names, existing or fictive. Once a decision has been made, the actual encoding in MLA should be straight forward, though.

7. A final word on the workflow

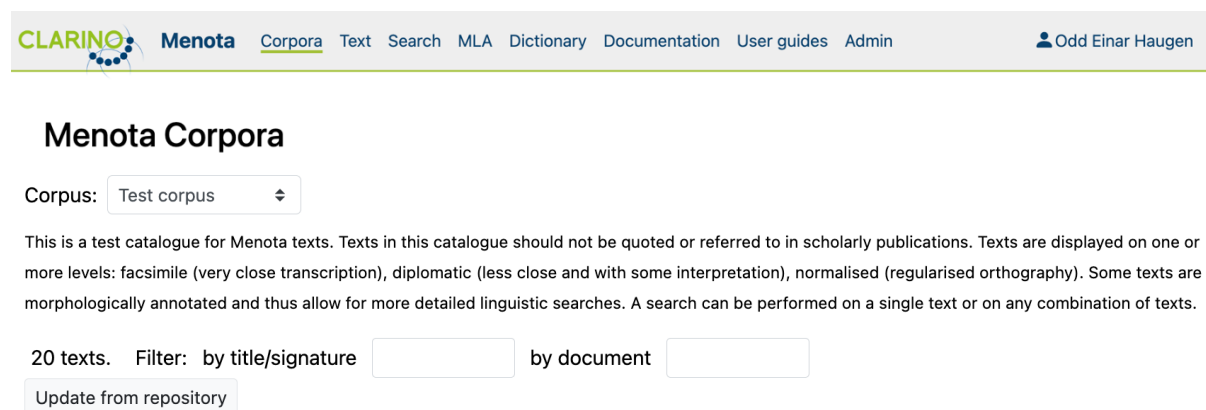
As stated in ch. 1 above, when working with a text, you should expect to switch between MLA and the downloaded version in XML. When downloading a text, go to **Text** in the menu bar and press **Download XML**:



After having downloaded a text in XML and edited it in any way you like, it must be uploaded to the Menota repository if you would like to see it on screen or continue working on it. For the time being, you may contact **odd.haugen@uib.no** or **paul.meurer@uib.no** for the uploading of texts in the Menota repository, while we are planning how to administer the necessary rights.

NB! Please check this User Guide to see any updates on the above contact information.

Assuming you have the necessary rights, go to **Corpora** in the menu bar and press **Update from repository**. It may take a few minutes for the update to be effected.



Remember that if a normalised level has been added, this must be stated in the **<normalization>** element of the header:

```
<normalization me:level="dipl norm">
  <p>This text has been encoded on a diplomatic and normalised level.</p>
</normalization>
```

Otherwise, the normalised level will not be displayed in the online Menota archive.

Appendix 1: Abbreviations of word classes and morphological categories

A. Word Classes

xNC	noun, common
xNP	noun, proper
xAJ	adjective
xPE	pronoun, personal
xPR	pronoun, reflexive
xPQ	pronoun, interrogative
xPI	pronoun, indefinite
xDD	determiner, demonstrative
xDQ	determiner, quantifier
xDP	determiner, possessive
xVB	verb
xAV	adverb, general
xAQ	adverb, interrogative
xAP	preposition
xCC	conjunction
xCS	subjunction
xIM	infinitive marker
xIT	interjection
xUA	unassigned
xFW	foreign word
xEY	empty (for Roman numerals and punctuation marks)

B. Morphological Categories

p1	person: 1. person
p2	person: 2. person
p3	person: 3. person
pU	person: unspecified
nS	number: singular
nP	number: plural
nU	number: unspecified
tPS	tense: present
tPT	tense: preterite
tU	tense: unspecified
mIN	mood: indicative
mSU	mood: subjunctive
mIP	mood: imperative
mINSU	mood: indicative or subjunctive
mINIM	mood: indicative or imperative

mSUIM	mood: subjunctive or imperative
mU	mood: unspecified
fF	finiteness: finite
fI	finiteness: infinitive
fP	finiteness: participle
fU	finiteness: unspecified
vA	voice: active
vR	voice: reflexive
vU	voice: unspecified
gM	gender: masculine
gF	gender: feminine
gN	gender: neuter
gMF	gender: masculine or feminine
gMN	gender: masculine or neuter
gFN	gender: feminine or neuter
gMFN	gender: masculine, feminine or neuter
gU	gender: unspecified
cN	case: nominative
cG	case: genitive
cD	case: dative
cA	case: accusative
cAD	case: accusative or dative
cGD	case: genitive or dative
cAN	case: accusative or nominative
cO	case: oblique
cU	case: unspecified
rP	degree: positive
rC	degree: comparative
rS	degree: superlative
rU	degree: unspecified
sl	definiteness: indefinite
sD	definiteness: definite
sU	definiteness: unspecified

Appendix 2: Keyboard shortcuts

Keyboard shortcuts on the MLA page

When *Key shortcuts w/o Ctrl* is checked, the Ctrl key does not have to be held down.

- Ctrl-Arrow-down: go to next reading or token
- Ctrl-Arrow-up: go to previous reading or token
- Ctrl-Arrow-right: go from token to reading. Opens all readings when *Collapse readings* is selected.
- Ctrl-Arrow-left: go from reading to token
- Ctrl-Shift-Arrow-right: go to next page
- Ctrl-Shift-Arrow-left: go to previous page
- Ctrl-K: open comment
- Ctrl-L: center reading or token on the page
- Ctrl-M: open Feature editor
- Ctrl-R: redisambiguate page (using Constraint Grammar rules)
- Ctrl-S: when in token mode: store proposed norm; when in comment box: store comment (needs Ctrl)
- Ctrl-X: when in reading mode, remove reading here and from database
- Ctrl-Y: open split token dialog. If token is already split, open undo split token dialog
- Ctrl-J: join this token with next token. If token is already joined, undo join
- Ctrl-P: add punctuation after this token
- Ctrl-Return: when in reading mode, select reading
- Ctrl-Space: when in reading mode, show all readings, including readings discarded by CG
- Escape: when in comment box: discard changes

Key shortcuts in the Feature editor

Here, the Ctrl key has to be held down when selecting a shortcut.

- Ctrl-Arrow-right: go to next feature
- Ctrl-Arrow-left: go to previous feature
- Ctrl-Arrow-down: go to next feature group
- Ctrl-Arrow-up: go to previous feature group
- Ctrl-S: store lemma, features, norm
- Ctrl-Return: select/deselect feature in focus
- Escape: discard changes